

Ekta

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EXPERIENCE

ResearchGapAI: Intelligent System for Research Paper Retrieval and Automated Research Gap Analysis

Research Project, NSUT

- Developed an end-to-end NLP system integrating Semantic Scholar, arXiv, Sentence Transformers, FAISS, and Llama-based LLMs.
- Implemented semantic retrieval and similarity search for context-aware research paper discovery.
- Automated identification of research gaps, research themes, and future research questions using Large Language Models.

Multimodal Rumor Detection on the PHEME Dataset using Deep Learning

Research Project, NSUT

- Designed a multimodal misinformation detection framework combining textual and visual information from social media posts.
- Implemented and evaluated CNN, GRU-RNN, and Transformer-based architectures on the PHEME dataset.
- Achieved up to 89% classification accuracy while analyzing model robustness and generalization.
- Explored multimodal fusion techniques for large-scale rumor detection.

Social Education and Liberty Foundation (SELF)

Community Intern

- Contributed 80+ hours toward education initiatives and community welfare programs.

Pehchaan – The Street School

Social Media Marketing Intern

- Managed digital content and social media outreach activities.

PROJECTS

AI Medical Chatbot | Python | LangChain | HuggingFace | Streamlit | RAG

- Developed a Retrieval-Augmented Generation (RAG) chatbot for medical document question answering.
- Built document ingestion, chunking, embedding, and semantic retrieval pipelines.
- Generated context-aware responses using HuggingFace models and LangChain.

Resume–Job Description Matcher | Python | Flask | NLP | TF-IDF

- Developed an NLP-based application to evaluate resume-job similarity.
- Implemented TF-IDF vectorization and cosine similarity for candidate-job matching.
- Added automated parsing for PDF, DOCX, and TXT resumes.

EDUCATION

Netaji Subhas University of Technology (NSUT)

B.Tech in Computer Science and Artificial Intelligence | CGPA: 8.32/10

2023 – 2027

Delhi

A.V. International Public School

Class XII (CBSE): 89% | Class X (CBSE): 94%

Sirsa, Haryana

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML, CSS

Machine Learning AI: Scikit-Learn, TensorFlow, Keras, NLP, Generative AI, LLMs, LangChain, HuggingFace, FAISS, RAG

Data Science: Pandas, NumPy, Matplotlib

Tools Platforms: Git, GitHub, Jupyter Notebook, VS Code, Streamlit

Core Subjects: Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Machine Learning

ACHIEVEMENTS

- Secured 2nd Position in IEEE Machine Learning Bootcamp.
- Completed the GeeksforGeeks 160 Days Coding Challenge with a 160-day streak.
- Solved 400+ DSA problems across LeetCode, GeeksforGeeks, and Codeforces